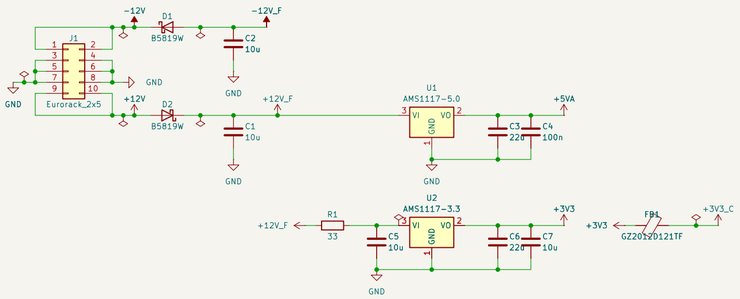
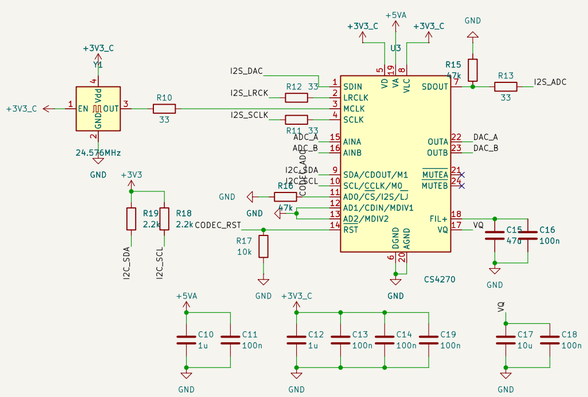
BIG MODULAR USB-AUDIO — carrier (jack/pot board), S1. 2 in / 2 out, DC-coupled +/-5 V, CS4270 24-bit 192 kHz. Circuit intent: docs/research.md; placement: docs/geometry.json

POWER: +12V -> D2 -> +12V_F (op-amps: U1 AMS1117-5.0 -> +5VA; R1 33R -> U2 AMS1117-3.3 -> +3V3 -> FB1 -> +3V3_C); -12V -> D1 -> -12V_F

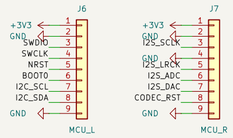


PWR_FLAG: GND/+12V/-12V at J1: +12V_F/-12V_F behind D2/D1; V3_IN behind R1; +3V3_C behind FB1.

CS4270 CODEC: I2C control port, address 0x48, Master, quad speed: MCLK 24.576 MHz / 128 = 192 kHz; SCLK = 64 x Fs to both CH32V307 I2S slaves (I2S2 RX, I2S3 TX), HPF freeze (reg 04h b7/b6) for DC. Firmware: RST high after clocks, write PDN within 1045 LRCK periods, then FM = quad master.

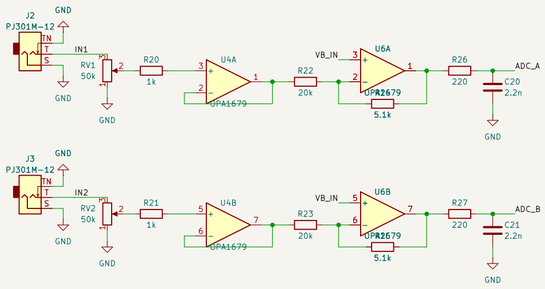


MCU BOARD LINK: J6/J7 1x9 2.54 mm male headers (plastic on the carrier F side); the MCU board (usb-audio-mcu) solders on top. BOOT/RESET and SWD on the carrier (B side): hold BOOT, press RESET = CH32V307 ROM bootloader on the USB-C.

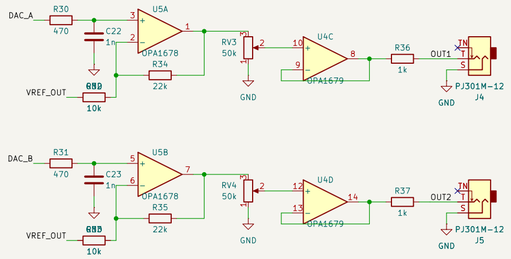




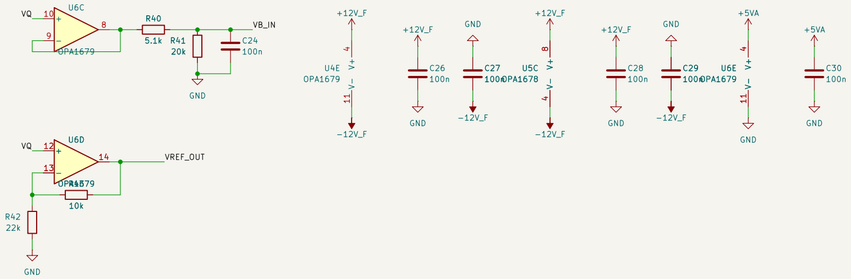
INPUTS: jack (TN = GND) -> 50k volume (0..1) -> 1k -> U4A/B follower (+/-12 V) -> U6A/B inverting driver on +5VA (~0.255, + at VB_IN) -> 220R / 2.2n C0G -> AINA/AINB. +/-5.49 V = 2.8 Vpp ADC full scale. ADC_INV restores the polarity.



OUTPUTS: AOUTA/B -> 470R / 1n C0G -> U5A/B gain 3.2 (Rg 10k to VREF_OUT, Rf 22k) -> 50k volume (0..1) -> U4C/D follower -> 1k -> jack. 3.25 Vpp DAC full scale -> +/-5.2 V. Power-up: AOUT starts at 0 V (Popguard) so the outputs sit near -8 V for ~0.4 s.



REFERENCES (+5VA): U6C buffers VQ (VA/2) -> VB_IN = VQ x 20/25.1 (input drivers); U6D: VREF_OUT = VQ x 32/22 (output stages). Both track the codec mid-rail.



Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S1 — 2 in / 2 out DC-coupled USB audio, CS4270 codec (192 kHz), +/-5 V, 6HP; MCU
Big Modular
Sheet: /
File: usb-audio.kicad_sch
Title: Big Modular — USB-AUDIO carrier (jack/pot board)
Size: A2 | Date: | **Rev: S1**
KiCad E.D.A. 10.0.6 | Id: 1/1